

M708 HW 06

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Exercise 3

For free, we have the diagram

$$\begin{array}{ccccccc}
 & 0 & & 0 & & 0 & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H \longrightarrow 0 \\
 & & \downarrow & & & & \downarrow \\
 0 & \longrightarrow & I^0 & \longrightarrow & I^0 \oplus K^0 & \longrightarrow & K^0 \longrightarrow 0 \\
 & & \downarrow & & & & \downarrow \\
 0 & \longrightarrow & I^1 & \longrightarrow & I^1 \oplus K^1 & \longrightarrow & K^1 \longrightarrow 0 \\
 & & \downarrow & & & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

It remains to construct the vertical maps and check exactness. We must give maps $G \rightarrow I^0$ and $G \rightarrow K^0$. The latter we already have, it's the composition $G \rightarrow H \rightarrow K^0$. The former is induced by $F \rightarrow I^0$, since I^0 is an injective object and $F \rightarrow G$ is an injection, so we can lift along it. We check that this is injective: the snake lemma tells us that we get a LES starting with the kernels of the three maps, followed by their cokernels, and the first and third maps are injective, so the second kernel (of the map we constructed) is 0 as well. Now, we are in the following situation:

$$\begin{array}{ccccccc}
 & 0 & & 0 & & 0 & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & F & \longrightarrow & G & \longrightarrow & H \longrightarrow 0 \\
 & & \downarrow \varphi^0 & & \downarrow \chi^0 & & \downarrow \psi^0 \\
 0 & \longrightarrow & I^0 & \longrightarrow & I^0 \oplus K^0 & \longrightarrow & K^0 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \text{coker } \varphi^0 & \longrightarrow & \text{coker } \chi^0 & \longrightarrow & \text{coker } \psi^0 \longrightarrow 0 \\
 & & \downarrow & & & & \downarrow \\
 0 & \longrightarrow & I^1 & \longrightarrow & I^1 \oplus K^1 & \longrightarrow & K^1 \longrightarrow 0 \\
 & & \downarrow & & & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

where the maps out of the cokernels are injective (by exactness), and we just need to produce an injective map $\text{coker } \chi^0 \rightarrow I^1 \oplus K^1$. This is exactly the same situation as when we started, so we just repeat this construction ad infinitum. By induction, we conclude that the desired SES of resolutions exists.

Exercise 4

This proof is basically the one in Voisin's book. Let $\mathcal{I} = [U \mapsto \bigoplus_{x \in U} I_x]$ where I_x is an injective abelian group containing $\mathcal{F}_x$ (for example) be an injective and flasque sheaf with an inclusion $\mathcal{F} \hookrightarrow \mathcal{I}$. Then we have a short exact sequence

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{G} \rightarrow 0$$

of sheaves on X . We claim that $\mathcal{I}(U) \rightarrow \mathcal{G}(U)$ is surjective for every open $U \subseteq X$, so let $\sigma \in \mathcal{G}(U)$. Since $\mathcal{I} \rightarrow \mathcal{G}$ is surjective stalkwise, we can lift σ to $\mathcal{I}$ locally, but it's not guaranteed a priori that these lifts will agree. However, if we have $V, W \subseteq U$ open and lifts $\tau_V \in \mathcal{I}(V)$, $\tau_W \in \mathcal{I}(W)$, then we can correct this by noting that $\tau_V|_{V \cap W} - \tau_W|_{V \cap W} \in \mathcal{F}(V \cap W)$ by exactness, then using flasqueness of $\mathcal{F}$ we extend it on V to χ_V , then returning to $\mathcal{I}$ we consider $\tau_V - \chi_V$. This is a section which restricts to τ_W on $V \cap W$, so it glues with it to a section $\tau \in \mathcal{I}(V \cup W)$, and by construction of $\mathcal{G}$, χ_V vanishes in $\mathcal{G}$, so it maps to $\sigma|_{V \cup W}$ in $\mathcal{G}$. Clearly then, we can keep extending our lift via this process to all of U (not inductive; we're just choosing a maximal lift). Thus $\mathcal{I} \rightarrow \mathcal{G}$ is surjective.

This implies in particular that $\mathcal{G}$ is flasque, since $\mathcal{I}$ is flasque, so for $V \subseteq U$, we can lift $\sigma \in \mathcal{G}(V)$ to $\mathcal{I}(V)$ then to $\mathcal{I}(U)$, then obtain a section in $\mathcal{G}(U)$ restricting to σ .

Since $\mathcal{I}$ has an injective resolution consisting of itself followed by zeros, we have $H^i(X, \mathcal{I}) = 0$ for $i \geq 1$. Then, the long exact sequence

$$0 \rightarrow \mathcal{F}(X) \rightarrow \mathcal{I}(X) \rightarrow \mathcal{G}(X) \rightarrow H^1(X, \mathcal{F}) \rightarrow H^1(X, \mathcal{I}) \rightarrow H^1(X, \mathcal{G}) \rightarrow H^2(X, \mathcal{F}) \rightarrow \dots$$

implies that $H^{i+1}(X, \mathcal{F}) = H^i(X, \mathcal{G})$ for $i \geq 1$, and $H^1(X, \mathcal{F}) = 0$ since we showed $\mathcal{I}(X) \rightarrow \mathcal{G}(X)$ is surjective. Since $\mathcal{G}$ is also flasque, we have that $H^1(X, \mathcal{G}) = 0$ too, so by induction, we conclude that $R^i\Gamma(\mathcal{F}) = H^i(X, \mathcal{F}) = 0$ for all $i > 0$.

Exercise 5

Suppose F has a left adjoint $G: \mathcal{B} \rightarrow \mathcal{A}$, and let $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$ be a short exact sequence in $\mathcal{A}$. We aim to show that $0 \rightarrow F(X) \rightarrow F(Y) \rightarrow F(Z)$ is exact in $\mathcal{B}$. It's well-known and easy to show that functors with a left adjoint preserve limits. Since kernels are limits, we know that $F(\ker(A \rightarrow B)) = \ker(F(A) \rightarrow F(B))$. Thus, we know exactness at the first position. Furthermore, since $F(X) \rightarrow F(Y)$ is an injection, we have $\text{im}(F(X) \rightarrow F(Y)) \cong F(X)$, and similarly we know $F(X) \cong F(\text{im}(X \rightarrow Y))$, so in this case F commutes with im , and we conclude exactness at the second position. Therefore, F is left-exact. The right-exactness statement is dual to the above argument.

Exercise 7

A space X is irreducible if and only if no two nonempty open sets are disjoint. If this is the case, clearly all open sets are connected, and hence every locally constant function on an open set is constant. Thus, any such section can be extended to a global section, so the constant sheaf is

flasque. Conversely, if X is not irreducible, then given a decomposition $X = A \cup B$ with A, B proper and closed, their complements are disjoint nonempty open sets, so a section taking distinct values on the two cannot extend to a global section (since X is connected, all such are constant).

Exercise 8

Garrett and I discussed some examples. Fix your favorite non-zero abelian group A . The two-point topological space with one generic point $\{x, y\}$ with open sets $\emptyset \subseteq \{x\} \subseteq \{x, y\}$ carries a non-flasque sheaf assigning A to $\{x\}$ and 0 to $\{x, y\}$. (Note that all presheaves are sheaves here, since any collection of open sets contains its union.) We can realize it as the quotient of the constant sheaf with stalk A (assigning $\{x\}, \{x, y\} \mapsto A$) by the sheaf assigning $\{x\} \mapsto 0, \{x, y\} \mapsto A$. Both of these are clearly flasque. More generally, given any irreducible variety X and irreducible subvariety $i: Y \hookrightarrow X$ with the Zariski topology, the quotient of the constant sheaf $\underline{A}_X$ by the pushforward of the restriction $i_*(i^{-1}(\underline{A}_X))$ is usually not flasque, having zero global sections despite having nonzero local sections (on the complement of Y).

Exercise 9

Let $V \subseteq U \subseteq X$ be open, and let $s \in \mathcal{F}(V)$; we want to extend s to a section in $\mathcal{F}(U)$. Let $\mathbb{Z}|_U$ be the sheafification of the sheaf on X defined by

$$W \mapsto \begin{cases} \mathbb{Z}^W & W \subseteq U \\ 0 & W \not\subseteq U \end{cases}$$

and similarly for V . We have a map of sheaves $\mathbb{Z}_V \rightarrow \mathcal{F}$ given on open sets $W \subseteq V$ by $f \mapsto f \cdot s|_W$. We also have an injective map of sheaves $\mathbb{Z}_V \rightarrow \mathbb{Z}_U$ induced by inclusion (clearly injective on stalks). Thus, since $\mathcal{F}$ is an injective sheaf, we obtain a map of sheaves $\mathbb{Z}_U \rightarrow \mathcal{F}$ extending $\mathbb{Z}_V \rightarrow \mathcal{F}$, and in particular, the constant section $1 \in \mathbb{Z}_U(U)$ maps to some section $t \in \mathcal{F}(U)$ which clearly restricts to the image of $1 \in \mathbb{Z}_V(V)$, which is $s \in \mathcal{F}(V)$. Therefore, we have our extension of s , so $\mathcal{F}$ is flasque.

Exercise 11

a) Let $f: X \rightarrow Y$ be continuous, let $\mathcal{F}$ be a sheaf on X , and let $f_*\mathcal{F}$ be its pushforward on Y . Let $U \subseteq Y$ be open, and let $\sigma \in f_*\mathcal{F}(U) = \mathcal{F}(f^{-1}U)$. Since $\mathcal{F}$ is flasque, we can extend σ to $\tau \in \mathcal{F}(X) = \mathcal{F}(f^{-1}Y) = f_*\mathcal{F}(Y)$ with $\tau|_U = \sigma$. Therefore, $f_*\mathcal{F}$ is flasque.

b) Take a flasque resolution $\mathcal{F} \rightarrow M^\bullet$ of $\mathcal{F}$. Note that for $Y \subseteq X$ a closed subspace, i_* is an exact functor: for an exact sequence of sheaves over Y , the stalks of their pushforwards over a point $x \in X$ are either all 0 ($x \notin Y$) since any open neighborhood $x \in U \subseteq X \setminus Y$ has 0 space of sections, or just the stalk over Y ($x \in Y$) by definition of the subspace topology, in which case we already know stalkwise exactness. Thus, $i_*\mathcal{F} \rightarrow i_*M^\bullet$ is a resolution of $i_*\mathcal{F}$, and it's still flasque by (a). Since $i_*M^k(X) = M^k(i^{-1}X) = M^k(Y)$, the complex we have whose cohomology is $H^k(X, i_*\mathcal{F})$ is equal to the one whose cohomology is $H^k(Y, \mathcal{F})$, so we conclude that $H^k(Y, \mathcal{F}) \cong H^k(X, i_*\mathcal{F})$. (Note that we know that these flasque resolutions compute the relevant cohomologies by exercise 4 and proposition 4.32 in Voisin.)

Exercise 12

a) We know that $\mathcal{E}xt^0(\mathcal{O}_X, \mathcal{G}) = \mathcal{G}$ by a general property of derived functors. Now, note that we have a natural isomorphism from the identity functor on the category of $\mathcal{O}_X$ -modules to $\mathcal{H}om(\mathcal{O}_X, -)$,

with component at $\mathcal{G}$ given by the map of sheaves $\mathcal{G} \rightarrow \mathcal{H}om(\mathcal{O}_X, \mathcal{G})$ defined over $U \subseteq X$ by $s \mapsto [1|_U \mapsto s|_U]$ (extended to all of $\mathcal{O}_X(U)$ by $\mathcal{O}_X(U)$ -linearity) for $s \in \mathcal{G}(U)$. This has an inverse given by $(F(U): \mathcal{O}_X(U) \rightarrow \mathcal{G}(U)) \mapsto F(U)(1|_U)$. The two compositions are clearly the identities, and this whole thing is clearly natural. Thus, $\mathcal{H}om(\mathcal{O}_X, -)$ is isomorphic to the identity functor, which is exact, so all its higher derived functors vanish (by the SES→LES property of derived functors), that is, $\mathcal{E}xt^i(\mathcal{O}_X, \mathcal{G}) = 0$ for $i > 0$.

b) The same isomorphism as in part (a), restricted to the open subset $U = X$, yields a natural isomorphism $\mathcal{H}om(\mathcal{O}_X, -) \Rightarrow \Gamma$. Therefore, the conclusion is immediate: $\text{Ext}^i(\mathcal{O}_X, \mathcal{G}) \cong H^i(X, \mathcal{G})$.

c) Pick an injective resolution $\mathcal{G} \rightarrow I^\bullet$. Since each I^i is injective, $\mathcal{H}om(-, I^i)$ is exact (and contravariant, recall) and the maps $I^i \rightarrow I^{i+1}$ induce natural transformations $\mathcal{H}om(-, I^i) \Rightarrow \mathcal{H}om(-, I^{i+1})$ functorially, so we obtain an exact sequence of complexes

$$0 \rightarrow \mathcal{H}om(\mathcal{F}'', I^\bullet) \rightarrow \mathcal{H}om(\mathcal{F}, I^\bullet) \rightarrow \mathcal{H}om(\mathcal{F}', I^\bullet) \rightarrow 0$$

Now, it's a general fact of homological algebra that short exact sequences of complexes induce long exact sequences in cohomology, and from Voisin we know that the right derived functors (Ext^i in this case) can be computed as the cohomology of the image of an injective resolution, so we conclude that there is a long exact sequence

$$\cdots \rightarrow \text{Ext}^i(\mathcal{F}'', \mathcal{G}) \rightarrow \text{Ext}^i(\mathcal{F}, \mathcal{G}) \rightarrow \text{Ext}^i(\mathcal{F}', \mathcal{G}) \rightarrow \text{Ext}^{i+1}(\mathcal{F}'', \mathcal{G}) \rightarrow \cdots$$

Furthermore, it is clear that everything in this construction commutes with restriction of sheaves, meaning it extends to the sections of $\mathcal{H}om(\mathcal{F}^*, \mathcal{G})$ over every open subset $U \subseteq X$ in a manner commuting with the restriction maps. Therefore, we conclude the long exact sequence

$$\cdots \rightarrow \mathcal{E}xt^i(\mathcal{F}'', \mathcal{G}) \rightarrow \mathcal{E}xt^i(\mathcal{F}, \mathcal{G}) \rightarrow \mathcal{E}xt^i(\mathcal{F}', \mathcal{G}) \rightarrow \mathcal{E}xt^{i+1}(\mathcal{F}'', \mathcal{G}) \rightarrow \cdots$$

as desired.

Exercise 13

a) We have a natural isomorphism $\mathcal{H}om_{\mathbb{Z}[G]}(\mathbb{Z}, -) \Rightarrow (-)^G$, whose component at a G -module A is as follows. For $n \in \mathbb{Z}$ and $g \in G$, since $g \cdot n = n$, we have $g \cdot f(n) = f(g \cdot n) = f(n)$, so any G -module homomorphism $\mathbb{Z} \rightarrow A$ lands in A^G , and conversely, since $\mathbb{Z}$ is cyclic and free, we can map $1 \in \mathbb{Z}$ to any chosen element of A^G . Clearly then, G -module homomorphisms $\mathbb{Z} \rightarrow A$ are in bijection with A^G , and this bijection is clearly natural in A . It follows then that the right derived functors of the two functors are isomorphic.

b) First, instead of writing $\mathbb{Z}[G]$ for $\mathbb{Z}[\mathbb{Z}]$ where $G = \mathbb{Z}$, I'll write $\mathbb{Z}[x, x^{-1}]$, since that ring is clearly isomorphic to the group ring in question. Note that we have a short exact sequence of $\mathbb{Z}[x, x^{-1}]$ -modules

$$0 \rightarrow \mathbb{Z}[x, x^{-1}] \rightarrow \mathbb{Z}[x, x^{-1}] \rightarrow \mathbb{Z} \rightarrow 0$$

where the first map is given by multiplication by $x - 1$ (clearly injective), and the second map is the quotient by the ideal $(x - 1) \subseteq \mathbb{Z}[x, x^{-1}]$ (clearly surjective onto $\mathbb{Z}$). Now note that the reasoning in exercise 12 above works for any ringed space, not just $\mathcal{O}_X$ -modules, and typical modules over rings are just the case of this over the singleton space, so by part (c), we have a long exact sequence

$$\cdots \rightarrow \text{Ext}^i(\mathbb{Z}, \mathbb{Z}) \rightarrow \text{Ext}^i(\mathbb{Z}[x, x^{-1}], \mathbb{Z}) \rightarrow \text{Ext}^i(\mathbb{Z}[x, x^{-1}], \mathbb{Z}) \rightarrow \text{Ext}^{i+1}(\mathbb{Z}, \mathbb{Z}) \rightarrow \cdots$$

Furthermore, by part (a), we have $\text{Ext}^i(\mathbb{Z}[x, x^{-1}], \mathbb{Z}) = 0$ for $i > 0$ and $\mathbb{Z}$ for $i = 0$. Thus, our LES looks like

$$\begin{aligned} 0 \rightarrow \text{Hom}_{\mathbb{Z}[x, x^{-1}]}(\mathbb{Z}, \mathbb{Z}) &\rightarrow \text{Hom}_{\mathbb{Z}[x, x^{-1}]}(\mathbb{Z}[x, x^{-1}], \mathbb{Z}) \rightarrow \text{Hom}_{\mathbb{Z}[x, x^{-1}]}(\mathbb{Z}[x, x^{-1}], \mathbb{Z}) \\ &\rightarrow \text{Ext}^1(\mathbb{Z}, \mathbb{Z}) \rightarrow 0 \rightarrow 0 \rightarrow \text{Ext}^2(\mathbb{Z}, \mathbb{Z}) \rightarrow 0 \rightarrow \dots \end{aligned}$$

Now note that the first three terms are isomorphic to $\mathbb{Z}$, and the map $\text{Hom}_{\mathbb{Z}[x, x^{-1}]}(\mathbb{Z}[x, x^{-1}], \mathbb{Z}) \rightarrow \text{Hom}_{\mathbb{Z}[x, x^{-1}]}(\mathbb{Z}[x, x^{-1}], \mathbb{Z})$ is given by precomposition with multiplication by $x - 1$, which is the zero map since x acts by 1 on $\mathbb{Z}$ with the trivial (group) $\mathbb{Z}$ -module structure. Thus, we conclude that $H^i(\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}$ for $i = 0, 1$, and $H^i(\mathbb{Z}, \mathbb{Z}) \cong 0$ for $i \geq 2$.